Chapter 7

Conclusion

Modern CMB experiments such as the Simons Observatory (SO) and CMB-S4 aim to investigate models of early universe inflation, constrain the sum of neutrino masses, probe the nature of dark energy and dark matter, and search for new relic particles beyond the Standard Model using instrumentation with unprecedented sensitivities. In this dissertation, we reviewed:

- the motivation behind this work and provided an overview of the dissertation
- a cosmological overview to contextualize the science goals and drivers
- an overview of modern CMB experiments and sensitivities, with a focus on SO
- a report of how a small aperture telescope for SO was integrated, tested, and deployed at the Chilean site, along with an overview of current and future work at the site
- a review of focal plane structures and developing detector arrays in CMB experiments, with an emphasis on technologies developed at Berkeley for SO along with CMB-S4 and *LiteBIRD*
- a summary on current and ongoing calibration and scan strategy work to get SO into a successful observation period

This work has successfully pushed one of the four telescopes (and the only ultra-high frequency small aperture telescope) for SO to deployment, developed and optimized detector arrays for current and future CMB experiments, and started building important frameworks for initial science observations with SO telescopes. This is an exciting time to be a part of the CMB community, and SO will be starting full science observations soon with much exciting development happening with upcoming experiments like CMB-S4!